ONLINE RETAIL DATASET — ONE-PAGE SUMMARY REPORT

Introduction

This project analyzed the Online Retail dataset obtained from Kaggle, which contains transactional data from a UK-based online retail store. The dataset includes invoice numbers, product stock codes, product descriptions, quantities purchased, invoice dates, unit prices, customer IDs, and customer countries. The aim of the analysis was to understand the dataset, clean it, explore patterns, and derive actionable business insights.

Cleaning Challenges Encountered

Several data quality issues were identified during the cleaning process. First, missing values were found in the Description and CustomerID columns. Missing product descriptions posed a challenge because they limited product-level analysis; these rows were removed. Missing customer IDs were handled by replacing them with “UNKNOWN” to preserve transaction records.

Duplicate records were also detected in the dataset. These duplicates could distort sales and revenue calculations, so they were removed to improve data accuracy.

Another challenge involved inconsistent formatting. Column names were standardized to lowercase for consistency, country names were converted to uppercase to avoid duplicate categories caused by case differences, and the InvoiceDate column was converted to datetime format to support time-series analysis.

Additionally, invalid records such as negative quantities and non-positive unit prices were identified. Negative quantities were interpreted as product returns or cancellations, while zero or negative prices were treated as anomalies and flagged during validation.

Key EDA Findings

Exploratory Data Analysis revealed several important patterns in customer purchasing behavior and sales performance.

First, product sales were highly concentrated among a small number of products. The top-selling products accounted for a significant share of total quantities sold, indicating strong demand for specific items.

Second, revenue generation was dominated by a few countries. The United Kingdom contributed the highest revenue by a large margin, showing that the business depends heavily on its domestic market.

Third, monthly sales trends showed noticeable fluctuations over time. Certain months recorded higher revenue, suggesting seasonal demand and peak shopping periods, especially around festive or holiday seasons.

Fourth, the unit price distribution showed that most products were sold at relatively low prices, while only a small number of products were premium-priced.

Finally, the quantity distribution revealed extreme outliers, indicating bulk purchases by some customers or large wholesale orders.

Top Insights
1. Sales concentration is high, with a small number of products driving most overall sales volume.
2. The United Kingdom is the dominant revenue source, making it the company's most important market.
3. Customer demand changes across months, showing clear seasonal purchasing behavior.
4. Most products are low-cost items, suggesting the store mainly serves price-sensitive buyers.
5. Bulk orders from certain customers significantly influence total revenue and sales volume.

Conclusion

The Online Retail dataset analysis provided valuable insights into sales performance, customer behavior, and revenue distribution. The cleaning process improved data quality and reliability, while EDA and visualization helped uncover meaningful trends. These findings can support better inventory planning, customer targeting, and strategic business decision-making.